



Customer Shopping Behavior Analysis

3,900 purchases — 18 features — goal: uncover spending patterns, segments, product & subscription insights

Dataset Summary

Scale

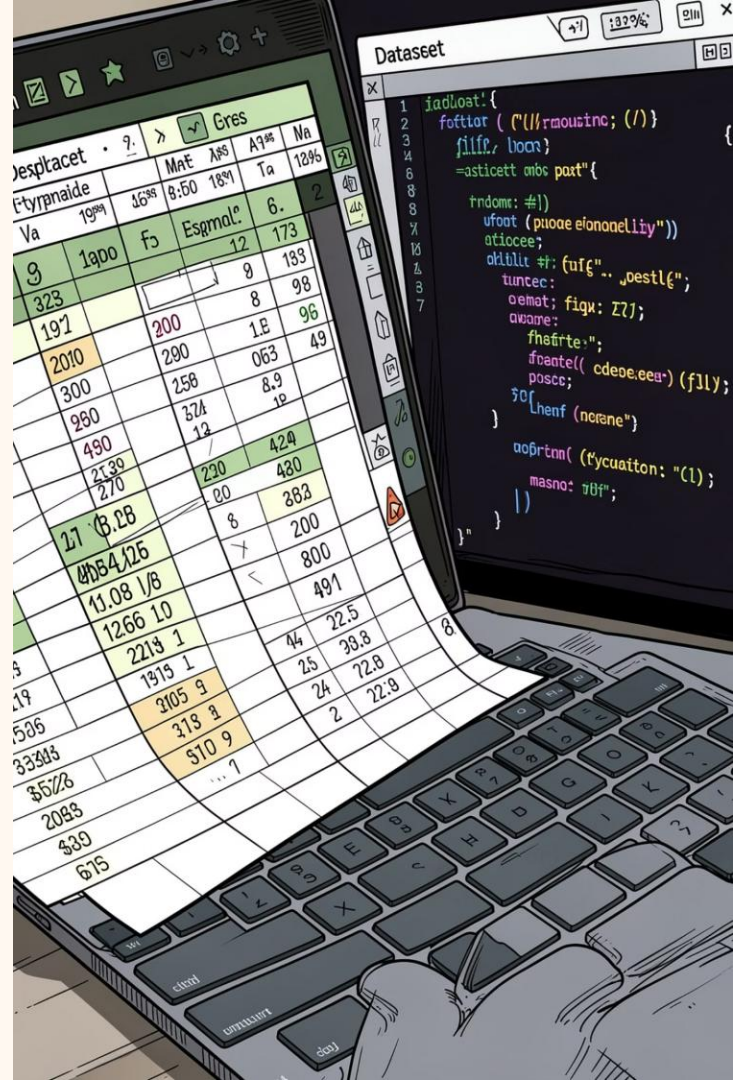
Rows: 3,900 · Columns: 18

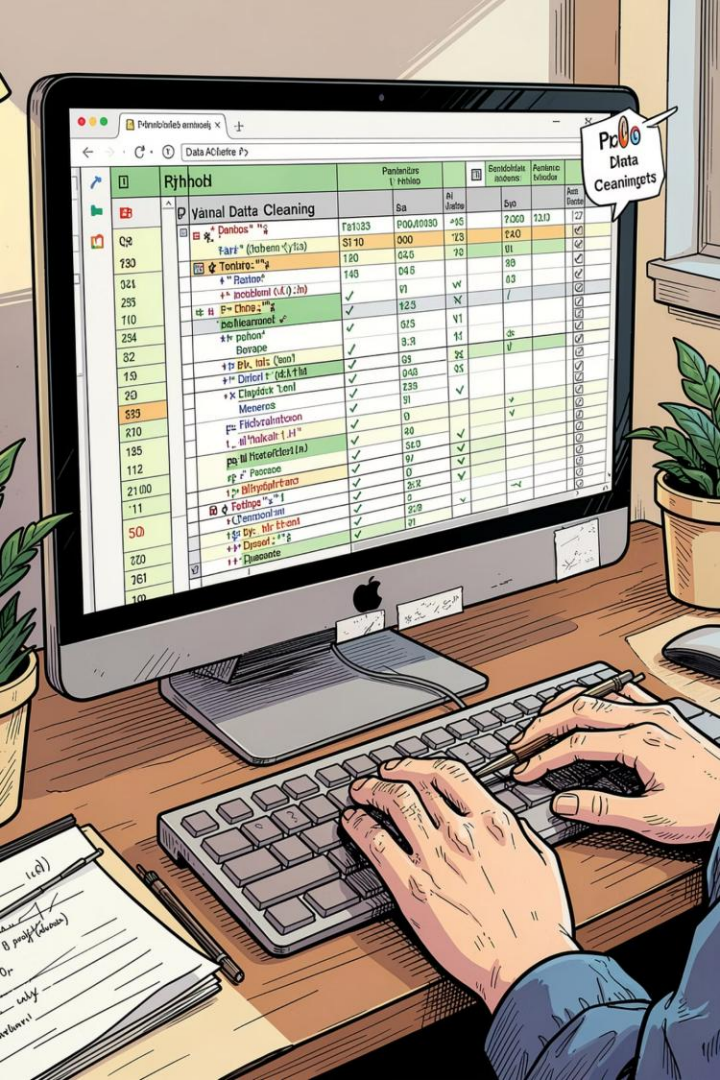
Key Features

Demographics, Purchase details, Behavior, Shipping, Discounts

Missing Data

37 nulls in Review Rating





Data Preparation & Feature Engineering



Loading & Explore

pandas: df.info(), describe()



Cleaning

Imputed Review Rating by category median



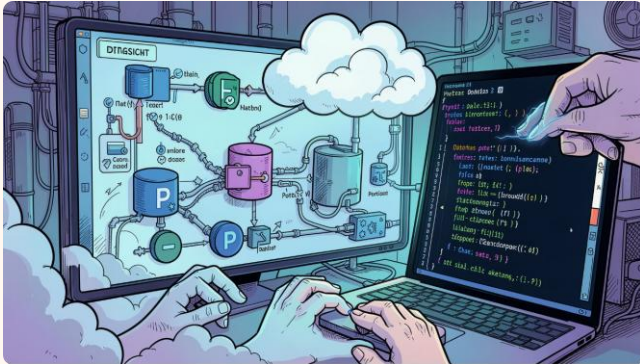
Features

age_group, purchase_frequency_days



Consistency

Dropped redundant promo_code_used



Database Integration

Cleaned DataFrame loaded into PostgreSQL for SQL analysis and reporting



Column names standardized to snake_case

Revenue Insights (SQL)



Revenue by Gender

Female: 75,191 · Male: 157,890



High-Spending Discount Users

839 rows identified — customers using discounts but spending above average



Product & Rating Insights

- Top rated items: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)
- Discount-dependent products: Hat 50%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37%

Shipping & Subscription Patterns

Shipping

Average spend:
Standard \$58.46 ·
Express \$60.48

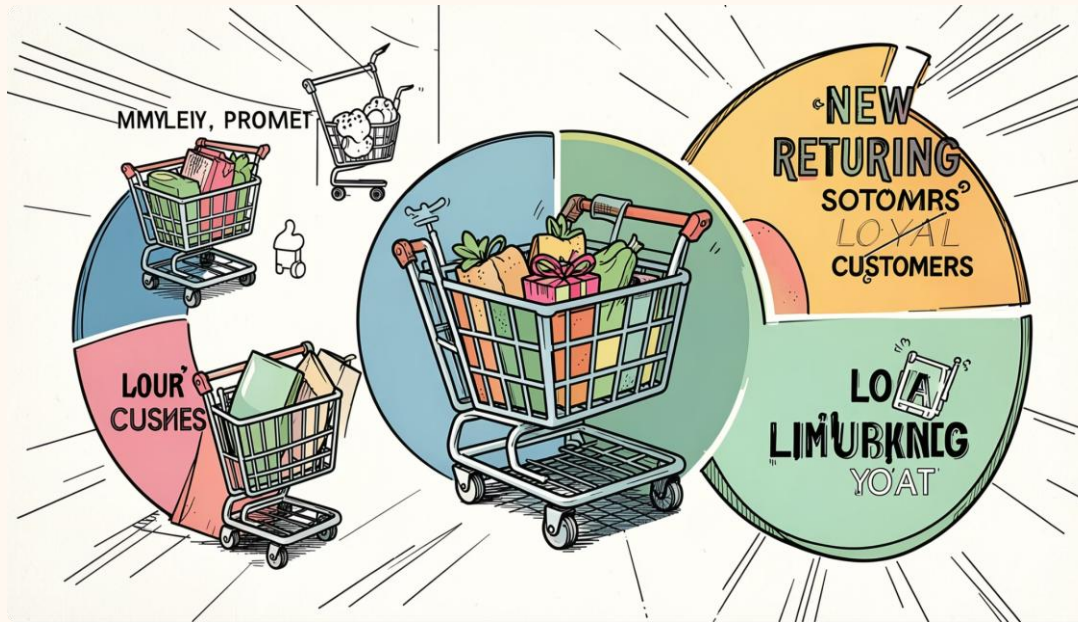
Subscriptions

Yes: 1,053
customers · Avg
spend \$59.49 ·
Total revenue
\$62,645

Non-Subscribers

No: 2,847
customers · Avg
spend \$59.87 ·
Total revenue
\$170,436





Customer Segmentation

Loyal: 3,116 · Returning: 701 · New: 83

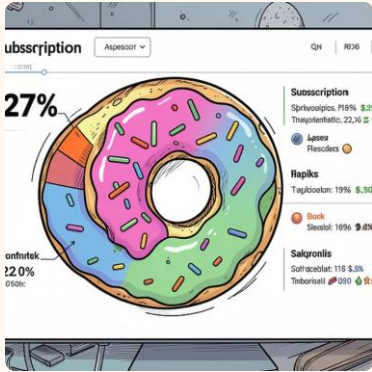
- Majority fall into Loyal segment — opportunity to retain and upsell

Repeat Buyers vs Subscriptions

Repeat buyers more likely to subscribe (958 vs 2,518 non-subscribed repeat buyers)

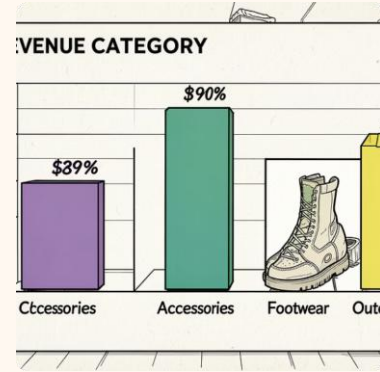
Power BI Dashboard

Interactive visuals: Subscription donut (Yes 27% / No 73%), Revenue by category, Sales by category, Revenue & Sales by age group



Key Metrics

3.9K customers · Avg purchase \$59.76 · Avg rating 3.75



Category Highlights

Revenue: Clothing ~100K · Accessories ~50K · Footwear ~25K
· Outerwear ~10K

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to increase loyalty

Review Discount Policy

Balance sales lift with margin control

Product Positioning

Feature top-rated & best-selling items in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users